

Developer Mock Test Results

Test ID:	fc2b1da-d63f-41...	Student ID:	24181
Test Type:	Developer Assessment	Date:	2026-03-04 14:50:21
Overall Score:	10/25 (40.0%)	Performance:	■ Average
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	4/10	40.0%	■■ Needs Work
■ MCQ/Theory	3/10	30.0%	■■ Needs Work
■ Coding	3/5	60.0%	■ Pass

Detailed Question Review

■ APTITUDE SECTION (4/10 - 40.0%)

■ Q1. If it is true that all books are written and some written things are articles, what can be concluded?

Your Answer: All articles are books

Correct Answer: Some written things are books

■ The correct answer is "Some written things are books" because we know that all books are written, which means books are a subset of written things. The user's answer, "All articles are books", is incorrect because the statement "some written things are articles" does not imply that all articles are books, only that there is an overlap between written things and articles. The user likely mistakenly assumed that the relationship between written things and articles is the same as the relationship between written things and books.

■ Q2. If 2 men can do a job in 6 days, how many days will it take for 1 man to do the same job?

Your Answer: 6

Correct Answer: 12

■ The correct answer is 12 because if 2 men can do the job in 6 days, it means that the total work required is 2 men * 6 days = 12 man-days. To find out how many days it will take for 1 man to do the job, we divide the total work by 1 man, which gives us 12 man-days / 1 man = 12 days. The user's mistake is likely due to not considering the total work required and the inverse relationship between the number of men and the number of days.

■ Q3. The average of 2 numbers is 18. What is the sum of the 2 numbers?

Your Answer: 36

■ Well done!

■ Q4. A mixture is made of 2 parts oil and 5 parts water. What is the ratio of oil to the total mixture?

Your Answer: 1/3

Correct Answer: 2/7

■ The correct answer is 2/7 because the total mixture is made of 2 parts oil and 5 parts water, making a total of $2 + 5 = 7$ parts. The ratio of oil to the total mixture is then 2/7. The user's likely mistake was calculating the ratio as $2/(2+5) = 2/7$, but possibly misinterpreting the fraction or calculating the total parts incorrectly, leading to their answer of 1/3.

■ **Q5. A mixture is made of 3 parts sugar and 2 parts water. What is the ratio of sugar to water?**

Your Answer: 3:1

Correct Answer: 3:2

■ The correct answer is 3:2 because the mixture is made of 3 parts sugar and 2 parts water, directly giving us the ratio of sugar to water. The user's answer, 3:1, is likely incorrect because it implies that for every 3 parts of sugar, there is only 1 part of water, which does not match the given proportions. The key calculation step is simply comparing the parts of sugar (3) to the parts of water (2), resulting in the ratio 3:2.

■ **Q6. What is the next number: 1, 2, 4, 8, 16, ?**

Your Answer: 32

■ Excellent! Right answer.

■ **Q7. A shopkeeper buys an item at Rs.200 and sells it at a 10% loss. What is the selling price?**

Your Answer: Rs.185

Correct Answer: Rs.190

■ The correct answer is Rs.180. The key calculation step is: $\text{Selling Price} = \text{Cost Price} - \text{Loss} = \text{Rs.200} - (10\% \text{ of Rs.200}) = \text{Rs.200} - \text{Rs.20} = \text{Rs.180}$. The user's likely mistake is calculating the loss amount incorrectly, as 10% of Rs.200 is Rs.20, not Rs.15, which would be the loss amount if the selling price were Rs.185.

■ **Q8. A shopkeeper buys an item at Rs.50 and sells it at a 15% profit. What is the selling price?**

Your Answer: Rs.57.50

■ Well done!

■ **Q9. If it is true that all dogs are animals and some animals are cats, what can be concluded?**

Your Answer: Some animals are dogs

■ Excellent! Right answer.

■ **Q10. If a number is increased by 10% and then decreased by 10%, what is the overall percentage change?**

Your Answer: 10%

Correct Answer: -1%

■ The correct answer is -1% because when a number is increased by 10% and then decreased by 10%, the overall effect is a decrease of 1%. The key calculation step is: if we start with 100, a 10% increase gives us 110, and then a 10% decrease gives us $110 - (0.1 * 110) = 110 - 11 = 99$, which is 1% less than the original 100. The user's likely mistake was assuming that the increase and decrease would cancel each other out, resulting in a 0% or 10% change, rather than calculating the actual effect of the consecutive percentage changes.

■ MCQ SECTION (3/10 - 30.0%)

■ **Q1. What is the purpose of a method that takes two numbers and returns their sum?**

Your Answer: To store data values

Correct Answer: To perform a specific task

■ The correct answer, "To perform a specific task", is right because a method that takes two numbers and returns their sum is designed to execute a particular operation, which is adding the two numbers together. The user's answer, "To store data values", is incorrect because storing data is typically the role of variables, not methods. The user likely mistook the purpose of a method, which is to perform an action or calculation, with that of a variable or data structure, which is to hold data.

■ **Q2. What type of block is used to handle runtime exceptions in Java?**

Your Answer: Try-catch block

■ Excellent! Right answer.

■ Q3. What is a good practice for keeping code organized in Java?

Your Answer: Using classes and packages

■ Correct! Good understanding.

■ Q4. What is the main focus of Java programming language?

Your Answer: None of the above

Correct Answer: Reliability, security, and portability

■ The correct answer is C) Reliability, security, and portability because Java programming language is designed with a focus on these three key aspects. Java's platform-independent nature, memory management, and built-in security features all contribute to its reliability, security, and portability. The user's answer, "None of the above", is likely incorrect because they may have overlooked or underestimated the importance of these core principles in Java's design philosophy.

■ Q5. What is the last step in writing a Java program?

Your Answer: Install Java (JDK) on your system

Correct Answer: Compile and run the program

■ The correct answer is "Compile and run the program" because this is the final step in writing a Java program, where the code is executed and tested for errors. The user's answer, "Install Java (JDK) on your system", is incorrect because it is a preliminary step that must be done before writing a Java program, not the last step. The user likely mistook a prerequisite for the final step in the programming process.

■ Q6. What are the types of control structures in Java?

Your Answer: Variables and data types

Correct Answer: Conditions and loops

■ The correct answer, "Conditions and loops", is right because control structures in Java refer to the ways in which the flow of a program's execution is managed. Conditions (if-else statements) and loops (for, while, do-while loops) are the primary control structures that dictate the order and repetition of statements in a Java program. The user's answer, "Variables and data types", is likely a mistake because these are fundamental components of Java programming, but they do not directly control the flow of a program's execution.

■ Q7. What are the types of data that can be stored in a variable in Java?

Your Answer: Int, double, char, String, arrays, etc.

■ Well done!

■ Q8. What are the prerequisites for installing Java?

Your Answer: A computer running Linux only

Correct Answer: A computer running Windows, macOS, or Linux

■ The correct answer is B) A computer running Windows, macOS, or Linux, because Java can be installed on multiple operating systems, including Windows, macOS, and Linux. The user's answer, C) A computer running Linux only, is incorrect because it limits the installation of Java to only Linux, excluding other compatible operating systems. The user's mistake is likely due to a lack of knowledge about Java's cross-platform compatibility.

■ Q9. What are control structures in Java?

Your Answer: Variables and data types

Correct Answer: Conditions and loops

■ The correct answer, "Conditions and loops", is right because control structures in Java refer to the elements that control the flow of a program's execution, such as if-else statements (conditions) and for/while loops. These structures determine the order in which code is executed, allowing for decision-making and repetition. The user's answer, "Variables and data types", is likely a mistake because variables and data types are fundamental components of Java, but they do not directly control the flow of a program's execution.

■ Q10. What is a good practice for testing programs in Java?

Your Answer: Test programs never

Correct Answer: Test programs regularly

■ The correct answer, "Test programs regularly", is the right choice because regular testing helps identify and fix bugs early on, ensuring the program's stability and reliability. This practice allows developers to catch errors and make necessary adjustments, ultimately saving time and resources. The user's answer, "Test programs never", is likely a mistake because it implies a lack of quality control, which can lead to faulty programs and dissatisfied users.

■ CODING SECTION (3/5 - 60.0%)

■ Q1. Write a program that reads a string and prints the string with all characters converted to uppercase.
Input Format: - Line 1: Input string (strin...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String text = sc.nextLine();

        System.out.println(text.toUpperCase());
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on writing robust and accurate code that meets the requirements.

■ Q2. Write a program that reads a string and an integer, and prints the result of dividing the integer by the length of the string. Handle any potential er...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String text = sc.nextLine();
        int num = sc.nextInt();

        try {
            double result = num / (double) text.length();
            System.out.println(result);
        } catch (ArithmeticException e) {
            System.out.println("Error: Division by zero");
        }
    }
}
```

Correct Answer:

```
def divide_integer_by_string_length():
    string = input()
    try:
        integer = int(input())
        if len(string) == 0:
            print("Error: Division by zero is not allowed")
        else:
            print(integer / len(string))
    except ValueError:
        print("Error: Invalid input")

divide_integer_by_string_length()
```

■ The student's code failed to handle the case where the input string is empty, causing a division by zero error. However, the catch block is expecting an `ArithmeticException`, which is not thrown in this case because the division is done using a `double`, avoiding the `ArithmeticException`. The code should handle the case where the string length is zero before performing the division.

■ **Q3. Write a program that reads a list of integers and prints the list with all negative numbers removed.**

****Input Format:** - Line 1: Number of integers (...)**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        for (int i = 0; i < n; i++) {
            int num = sc.nextInt();
            if (num >= 0) {
                System.out.print(num + " ");
            }
        }
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code that meets the requirements.

■ **Q4. Write a program that simulates a bank account, allowing users to deposit, withdraw, and check their balance. **Input Format:** - Line 1: Initial bala...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int balance = sc.nextInt();
        int transactions = sc.nextInt();

        for (int i = 0; i < transactions; i++) {
            String type = sc.next();
            int amount = sc.nextInt();

            if (type.equals("deposit")) {
                balance += amount;
            } else if (type.equals("withdraw")) {
                balance -= amount;
            }
        }

        System.out.println(balance);
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code!

■ **Q5. Write a program that reads a dictionary of word frequencies and prints the word with the highest frequency. **Input Format:** - Line 1: Number of wor...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        String maxWord = "";
        int maxFreq = -1;

        for (int i = 0; i < n; i++) {
            String word = sc.next();
            int freq = sc.nextInt();

            if (freq > maxFreq) {
                maxFreq = freq;
                maxWord = word;
            }
        }

        System.out.println(maxWord);
    }
}
```

Correct Answer:

```
n = int(input())
word_freq = {}
for _ in range(n):
    word, freq = input().split()
    word_freq[word] = int(freq)
print(max(word_freq, key=word_freq.get))
```

■ The code failed to handle cases where there are multiple words with the same highest frequency, as it only stores the first word it encounters with the maximum frequency. The output differs from the expected because it may not always print the correct word when there are multiple words with the same highest frequency. The code needs to be modified to handle such cases.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.